

A Multi-Modal Markov-Chain Passenger-Flow Model for the BKK Budapest GTFS Network: Entropy Maximisation, Kolmogorov Equations, and Stochastic Simulation*

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Abstract

We develop, implement, and critically evaluate a multi-modal Markov-chain passenger-flow model for the BKK Budapest public transit network, calibrated from the GTFS (General Transit Feed Specification) schedule. The model operates on three complementary stochastic layers: (L1) a discrete-time Markov chain (DTMC) for tractable analytical computation; (L2) a continuous-time Markov chain (CTMC) governed by the Kolmogorov Forward Equation, whose generator Q^v is derived from Wilson’s entropy-maximisation principle applied to the GTFS-derived generalised cost structure; and (L3) exact stochastic realisations via the Doob–Gillespie Stochastic Simulation Algorithm (SSA) with a τ -leaping approximation for large populations.

We make several contributions beyond the standard formulation: (i) a rigorous derivation of the CTMC generator from entropy maximisation with proper dimensional analysis; (ii) explicit handling of graph sparsity and disconnected origin–destination pairs; (iii) a hierarchical Poisson–log-linear prior for stop-level initial passenger counts, anchored to BKK’s published system ridership, in the absence of automated fare-collection data; (iv) a node-removal resilience analysis that explicitly distinguishes irreducible from reducible chains; and (v) a reproducible correction record. Previously reported stop rankings and KFE–SSA comparisons are withdrawn because audit revealed a dimensional rate error and incompatible modal state indices. The corrected implementation and regression tests are the primary contribution; new empirical claims require a pinned-feed rerun.

Limitations: The model is a first-principles construct without AFC/APC validation data. The gravity kernel imposes independence across passengers and modes; capacity constraints are absent from the base formulation; and the time-homogeneous CTMC is an idealisation of the time-varying GTFS schedule. These limitations are discussed quantitatively throughout.

Keywords: stochastic transit assignment, entropy maximisation, continuous-time Markov chain, Gillespie algorithm, GTFS, network resilience, Kemeny constant.

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1 Introduction

1.1 Motivation and Problem Setting

The Budapest metropolitan public transit network operated by Budapesti Közlekedési Központ (BKK) is one of Europe’s denser urban transit systems, serving on the order of 4×10^6 passenger journeys per weekday across six modal families (metro, bus, tram, trolleybus, suburban railway, and ferry). Despite this scale, spatially resolved passenger flow—the number of passengers at each stop at each point in time—is not directly observable from the publicly available schedule data alone.

The practical stakes are high. Spatial passenger density determines platform crowding risk, vehicle overcrowding, intermodal transfer delays, and the propagation of service disruptions. A principled model for passenger flow dynamics, even an imperfect one grounded in well-understood probabilistic machinery, enables:

1. **Network resilience assessment:** identifying stops and links whose failure disproportionately degrades service.
2. **Prior initialisation:** providing starting distributions for data-assimilation procedures once partial measurements (APC counters, AFC tap-in records) become available.
3. **Scenario analysis:** evaluating the system-level impact of new routes, closures, or timetable changes without requiring demand surveys.

1.2 Related Work

Entropy-based transport modelling. Wilson’s [24] doubly-constrained gravity model remains the theoretical backbone of spatial interaction modelling. Its maximum-entropy derivation has been formalised and extended by Erlander and Stewart [7] and connected to random utility models by Anas [2]. Recent work has embedded entropy transport models within the Wasserstein optimal-transport framework [19].

Stochastic transit assignment. The logit-based stochastic user equilibrium (SUE) model of Dial [6] and the frequency-based assignment of Spiess and Florian [21] are the industry standards. Dynamic stochastic assignment extending to time-varying supply is treated in Yildirimoglu and Kim [25] and the survey of Bliemer et al. [3]. None of these papers couple the assignment to a CTMC generator or uses Gillespie simulation.

CTMC models for transit. Markov-chain models have been applied to pedestrian flow [11], station waiting queues [13], and subway dwell times [26], but full-network CTMC formulations over GTFS-structured schedules are uncommon. The closest precedent is the queueing-network approach of Heidergott et al. [10].

Gillespie simulation beyond biochemistry. Although Gillespie’s SSA [8] was developed for chemical reaction networks, it applies to any CTMC. Applications in ecology [23], epidemiology [1], and traffic [17] demonstrate its scope.

Network resilience. The node-removal framework we adopt draws on Latora and Marchiori [15], Kemeny and Snell [14], and the transit-specific vulnerability analysis of Cats and Jenelius [5].

1.3 Contributions and Scope

C1. CTMC generator from GTFS. We show precisely how the Wilson entropy solution translates into a CTMC generator whose units are dimensionally consistent and whose entries are computable from `stop_times.txt`.

- C2. Honest regime analysis.** We document quantitatively the gap between the KFE mean-field trajectory and the finite-rate SSA trajectory, and explain when each is the appropriate model.
- C3. Sparse-graph formulation.** The full $n \times n$ cost matrix is never formed; all operations exploit the $\ll 1\%$ connectivity of the real GTFS graph.
- C4. Hierarchical demand prior.** A calibrated Poisson log-linear prior provides plausible initial passenger counts from freely available covariates.
- C5. Principled resilience ranking.** Three complementary vulnerability metrics (Kemeny, spectral gap, efficiency) are combined into a criticality score with $O(Kn^2)$ complexity.

1.4 Paper Organisation

Section 2 formalises the network graph. Section 3 derives the entropy-maximising flow and transition probabilities. Section 4 develops the CTMC generator and the Kolmogorov Forward Equation. Section 5 presents the hierarchical demand prior. Section 6 covers the Gillespie SSA and τ -leaping. Section 7 gives a computational complexity analysis. Section 8 describes the Python implementation. Section 9 reports empirical results on the full BKK network. Section 10 develops the resilience analysis. Section 11 is an honest discussion of limitations. Section 12 concludes.

2 Network Graph and GTFS Data Model

2.1 GTFS Inventory

Table 1: BKK GTFS files used in this work.

GTFS File	Key Fields Used	Role in Model
stops.txt	stop_id, stop_lat, stop_lon	Vertex set S , $ S = n$
routes.txt	route_id, route_type	Mode classification v
trips.txt	trip_id, route_id, service_id	Trip-route link
stop_times.txt	arrival_time, departure_time	Scheduled travel times $\bar{\tau}_{ij}^v$
shapes.txt	shape_pt_lat, shape_pt_lon	Route distance d_{ij}^v
calendar.txt	monday...friday	Weekday service filter
transfers.txt	from_stop_id, min_transfer_time	Transfer penalty δ_i

2.2 Multi-Modal Stop Graph

Definition 2.1 (Multi-modal stop graph). Let $S = \{1, \dots, n\}$ denote the set of GTFS stops and $\mathcal{V} = \{1, \dots, m\}$ the set of vehicle modes. For each mode $v \in \mathcal{V}$, define the modal subgraph $G^v = (S^v, E^v)$, where $S^v \subseteq S$ is the set of stops served by mode v and

$$E^v = \{(i, j) \in S^v \times S^v : \exists \text{ a trip of mode } v \text{ visiting } i \text{ immediately before } j\}.$$

The multi-modal graph is the union $G = (S, \bigcup_v E^v)$.

Remark 2.2 (Sparsity). On the BKK network $|E^v| \ll |S^v|^2$ for all modes. The bus sub-graph has $\approx 0.5\%$ fill. All matrices are stored in CSR format; the full $n \times n$ dense matrix is never materialised.

Definition 2.3 (Neighbourhood sets). $\mathcal{N}^v(i) = \{j \in S^v : (i, j) \in E^v\}$, $\mathcal{N}_{\text{in}}^v(i) = \{j \in S^v : (j, i) \in E^v\}$.

Table 2: BKK vehicle modes. κ^v is the mode-specific comfort weight.

v	Mode	rt	Speed (km/h)	Cap. (pax)	κ^v	Example Lines
1	Bus	3	18–25	80–180	1.00	7, 8E, 100E, 200E
2	Tram	0	15–22	200–350	0.70	2, 4, 6, 47–49
3	Trolleybus	800	15–20	100–170	0.85	70, 72, 74, 76
4	Metro	1	30–40	800–1000	0.40	M1, M2, M3, M4
5	Suburban rly	2	25–40	400–600	0.55	H5, H6, H7, H8
6	Ferry	4	12–18	100	1.20	D11, D12, D14

2.3 BKK Mode Classification

2.4 Admissible Transitions and Disconnected Pairs

A transition (i, j) on mode v is admissible iff $(i, j) \in E^v$. We assign a finite cost only to admissible transitions; all others receive $C_{ij}^v = +\infty$, yielding $\exp(-\lambda^v C_{ij}^v) = 0$ and zero probability/rate. This is a strictly correct handling of graph sparsity: we do not approximate disconnected pairs with a large finite penalty C_∞ , which would inject spuriously small but non-zero probabilities across the entire network.

Remark 2.4 (Self-loops). The entropy-maximisation formulation excludes self-loops ($i \neq j$ in the constraint set); accordingly C_{ii}^v plays no role. The correct correspondence is that P^v as defined in Definition 3.5 is the *embedded jump chain* of the CTMC, not the Δt -uniformisation.

3 Entropy Maximisation and the Softmax Flow

3.1 Generalised Cost Matrix

Definition 3.1 (Generalised cost). For admissible pair $(i, j) \in E^v$,

$$C_{ij}^v = \kappa^v \bar{\tau}_{ij}^v + \gamma d_{ij}^v + \delta \mathbf{1}_{\{i \in \mathcal{T}\}} t_{\text{tf}}(i), \quad (1)$$

where $\bar{\tau}_{ij}^v$ [s] is the median scheduled travel time, $\kappa^v > 0$ is the mode-specific comfort weight, d_{ij}^v [m] is the in-vehicle distance, γ [s m⁻¹] is the distance penalty, $t_{\text{tf}}(i)$ [s] is the minimum transfer time at stop i , and $\delta \in \{0, 1\}$ indicates a mode change. All terms are in seconds.

Remark 3.2 (Dimensional consistency). C_{ij}^v [s] and λ^v [s⁻¹] make the exponent $\lambda^v C_{ij}^v$ dimensionless, consistent with the interpretation of λ^v as an inverse temperature governing cost sensitivity.

3.2 Entropy-Maximising Flow

Assumption 3.3 (Separable origin constraint). Passengers are assigned a departure mode v before departure; within a mode, they choose destinations to maximise entropy subject to an expected-cost constraint.

Proposition 3.4 (Wilson softmax flow). *Under Assumption 3.3, the flow $F_{ij}^v \geq 0$ maximising Shannon entropy $H = -\sum_{v,i,j} F_{ij}^v \ln F_{ij}^v$ subject to the origin constraint $\sum_j F_{ij}^v = N_i^v(0)$ and the cost constraint $\sum_{ij} F_{ij}^v C_{ij}^v \leq \bar{C}^v$ is given by*

$$F_{ij}^v = N_i^v(0) \pi_{j|i}^v, \quad \pi_{j|i}^v = \frac{\exp(-\lambda^v C_{ij}^v)}{\sum_{k \in \mathcal{N}^v(i)} \exp(-\lambda^v C_{ik}^v)}, \quad (2)$$

where $\lambda^v \geq 0$ is the Lagrange multiplier, uniquely determined by the binding cost constraint.

Proof sketch. The Lagrangian yields $\ln F_{ij}^v = -1 + \mu_i^v - \lambda^v C_{ij}^v$, i.e. $F_{ij}^v \propto \exp(-\lambda^v C_{ij}^v)$. Row normalisation gives (2). Convexity of $-H$ ensures a unique maximiser. $\square$

Definition 3.5 (DTMC transition matrix). The row-stochastic DTMC transition matrix for mode v is $P_{ij}^v = \pi_{j|i}^v$ ($i \neq j$), $P_{ii}^v = 0$ (no self-loop), $P_{ij}^v = 0$ for $(i, j) \notin E^v$.

3.3 DTMC Time Evolution

The population vector on mode v at step t evolves as

$$N^v(t+1) = N^v(t) P^v, \quad N_i(t) = \sum_{v=1}^m N_i^v(t). \quad (3)$$

4 Continuous-Time Markov Chain and the Kolmogorov Forward Equation

4.1 Motivation for the CTMC Formulation

The GTFS records precise departure timestamps at each stop. Rather than discretising the schedule into fixed steps Δt , we embed the passenger movement process in continuous time. Under the memoryless (Markov) approximation for inter-departure intervals—justified when vehicles run at roughly constant headway—the waiting time is exponentially distributed and the process is a CTMC.

Assumption 4.1 (Poisson arrivals). The departure process at stop i on mode v is approximated as a Poisson process with rate f_i^v [vehicles s^{-1}] estimated from the GTFS timetable.

4.2 CTMC Generator

Definition 4.2 (Infinitesimal generator). For mode v , the generator $Q^v \in \mathbb{R}^{n \times n}$ is defined by

$$Q_{ij}^v = \begin{cases} q_{ij}^v \geq 0, & j \neq i, (i, j) \in E^v, \\ 0, & j \neq i, (i, j) \notin E^v, \\ -\sum_{k \in \mathcal{N}^v(i)} q_{ik}^v, & j = i. \end{cases} \quad (4)$$

All off-diagonal entries are non-negative and all row sums vanish.

4.3 Rates from Entropy Kernel and GTFS Frequencies

The off-diagonal rate decomposes as

$$q_{ij}^v = \underbrace{\mu_i^v}_{\text{departure intensity}} \cdot \underbrace{\pi_{j|i}^v}_{\text{Wilson kernel}}, \quad (i, j) \in E^v, j \neq i, \quad (5)$$

where the per-passenger departure hazard is the scheduled vehicle frequency

$$\mu_i^v = f_i^v \quad [s^{-1}]. \quad (6)$$

Here f_i^v [vehicles s^{-1}] is the peak-window departure frequency. This correction is dimensionally necessary: every entry of a CTMC generator has units s^{-1} . Capacity $\bar{k}^v$ and load factor ρ^v describe supply and demand, but multiplying them into Q and then again by N_i in the KFE/SSA propensity would produce the invalid unit $\text{pax}^2 s^{-1}$.

4.4 DTMC–CTMC Equivalence

Proposition 4.3 (Matrix exponential embedding). *The matrix exponential $P^v(\Delta t) := \exp(Q^v \Delta t)$ is stochastic for all $\Delta t \geq 0$; its first-order expansion recovers the Wilson DTMC entry structure with rescaled self-loop probabilities. The DTMC of Definition 3.5 is the embedded jump chain of the CTMC, observed at transition moments stripped of holding times.*

4.5 Kolmogorov Forward Equation

Theorem 4.4 (KFE). *Let $p^v(t) \in [0, 1]^n$ denote the normalised probability distribution of a single tagged passenger of mode v . Then*

$$\frac{dp^v(t)}{dt} = p^v(t) Q^v, \quad p^v(0) = p_0^v, \quad (7)$$

with unique solution $p^v(t) = p_0^v \exp(Q^v t)$.

By linearity and the mean-field (fluid-limit) approximation, the same equation holds for population counts:

$$\frac{dN^v(t)}{dt} = N^v(t) Q^v, \quad N^v(0) = N_0^v. \quad (8)$$

Conservation of mass follows from $Q^v \mathbf{1} = \mathbf{0}$.

4.6 Stationarity, Ergodicity, and Existence of Stationary Distributions

Assumption 4.5 (Irreducibility per mode). For each mode v , the directed graph G^v is strongly connected.

Proposition 4.6 (Ergodicity). *Under Assumption 4.5, Q^v is irreducible and the CTMC is ergodic. A unique stationary distribution π^v with $\pi^v Q^v = \mathbf{0}$, $\|\pi^v\|_1 = 1$ exists, and $p^v(t) \rightarrow \pi^v$ exponentially fast.*

Remark 4.7 (Irreducibility in practice). The BKK graph is not strongly connected for every mode in isolation. In implementation we partition S^v into strongly connected components using `scipy.sparse.csgraph` and analyse each separately.

Remark 4.8 (Time-homogeneity and GTFS schedules). A real GTFS schedule is time-varying; the time-homogeneous CTMC is valid only within a narrow service window (e.g., 07:00–09:00 peak hour) where f_i^v is approximately constant.

5 Prior Estimation of Initial Passenger Counts

5.1 Setting and Identifiability

In the absence of APC or AFC data, the initial count $N_i(0)$ is unobserved. We construct a prior $\hat{N}_i(0)$ from freely available covariates and calibrate its overall scale to BKK’s published aggregate ridership.

Assumption 5.1 (Log-linear demand).

$$N_i(0) \sim \text{Poisson}(\lambda_i), \quad \log \lambda_i = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i + \xi_i, \quad \xi_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \quad (9)$$

5.2 Feature Vector

Table 3: Features used in the log-linear demand prior.

Feature	Symbol	Source	Unit
Peak-hour service frequency	f_i	GTFS	trips h^{-1}
Population in 400–800 m catchment	Pop_i	KSH/WorldPop	persons
POI count in 400 m catchment	POI_i	OpenStreetMap	count
Network betweenness centrality	B_i	GTFS graph	[0,1]
Interchange indicator	I_i	GTFS transfers	{0, 1}
Distance to Deák F. tér	D_i	Haversine	km

Table 4: Prior distributions $\mathcal{N}(\mu_\beta, \sigma_\beta^2)$ for log-linear coefficients.

Covariate (log-transformed)	Symbol	μ_β	σ_β	Basis
$\log(1 + f_i)$	β_1	+0.70	0.10	Service elasticity [20]
$\log(1 + \text{Pop}_i)$	β_2	+0.60	0.15	Population elasticity
$\log(1 + \text{POI}_i)$	β_3	+0.20	0.10	Activity generation
B_i	β_4	+0.30	0.15	Network centrality
I_i	β_5	+0.50	0.20	Transfer-hub effect
$\log(1 + D_i)$	β_6	-0.25	0.10	Distance decay

5.3 Literature-Informed Prior Coefficients

5.4 Calibration Anchor

BKK reports $N_{\text{day}} \approx 4 \times 10^6$ weekday boardings. With $\phi_{\text{peak}} \approx 0.11$,

$$N_{\text{peak}} = \phi_{\text{peak}} N_{\text{day}} \approx 440,000 \text{ passengers.} \quad (10)$$

The intercept β_0 is set so that $\sum_i \hat{N}_i(0) = N_{\text{peak}}$.

5.5 Diurnal Profile

$\hat{N}_i(0; h) = 24 \phi(h) \hat{N}_i(0)$ with $\phi(07:00-09:00) \approx 0.11$.

5.6 Three Estimators

E1 — Service proxy (GTFS only). $\hat{N}_i(0) \propto f_i \cdot \bar{k}^{v_i} \cdot \rho$. No external data needed; captures frequency variation but ignores land use.

E2 — Poisson log-linear (recommended). Equation (9) with priors from Table 4.

E3 — Full Bayesian (for future use with AFC/APC data). Full posterior inference via MCMC. Reduces to E2 when no observations are available.

5.7 Uncertainty Propagation

Monte Carlo propagation: draw $\beta^{(s)} \sim \mathcal{N}(\mu_\beta, \Sigma_\beta)$, compute $\hat{N}_i^{(s)}(0)$, run KFE or SSA, report empirical quantiles.

6 Gillespie Stochastic Simulation Algorithm

6.1 Reaction-Channel Representation

The propensity of channel $(v, i \rightarrow j)$ is

$$a_{ij}^v(t) = N_i^v(t) q_{ij}^v, \quad a_0(t) = \sum_v \sum_i \sum_{j \in \mathcal{N}^v(i)} a_{ij}^v(t). \quad (11)$$

This is exact for $N_i^v(t) \in \mathbb{Z}_{\geq 0}$; the mean-field ODE (8) is recovered in the limit $N_i^v \rightarrow \infty$.

6.2 Direct Method (Exact SSA)

Implementation note (v2.1): Channel selection uses the $O(R)$ inverse-CDF method (`np.cumsum + np.searchsorted`), which is simpler than the Vose alias method [22] and sufficient at the BKK

event rates encountered in practice. The alias method remains an $O(1)$ alternative for very large R and is noted for future optimisation.

Algorithm 1: Gillespie Direct Method for BKK Multi-Modal CTMC

Require: Sparse channels $\{(v, i, j, q_{ij}^v)\}$ (only $(i, j) \in E^v$); initial state $N(0)$; horizon T .

Ensure: Event list $\mathcal{E}$; final state $N(T)$.

1. $t \leftarrow 0, \quad \mathcal{E} \leftarrow \emptyset$
2. **while** $t < T$ **do**
 - (a) Compute propensities (vectorised): $a[r] \leftarrow N_i^v \cdot q_{ij}^v; \quad a_0 \leftarrow \sum_r a[r]$
 - (b) **if** $a_0 = 0$ **then break**
 - (c) Draw $\tau \leftarrow -\ln(u_1)/a_0, \quad u_1 \sim \mathcal{U}(0, 1)$
 - (d) **if** $t + \tau > T$ **then break**
 - (e) Select r^* via inverse-CDF on $\{a[r]/a_0\}$ using $u_2 \sim \mathcal{U}(0, 1)$
 - (f) $t \leftarrow t + \tau; \quad N_{i^*}^{v^*} \leftarrow N_{i^*}^{v^*} - 1; \quad N_{j^*}^{v^*} \leftarrow N_{j^*}^{v^*} + 1$
 - (g) Append (t, v^*, i^*, j^*) to $\mathcal{E}$
3. **return** $\mathcal{E}, N(T)$

Figure 1: Gillespie Direct Method for BKK Multi-Modal CTMC.

6.3 τ -Leaping Approximation

When $N_i^v \gg 1$, the τ -leap [9] advances by a fixed step τ and draws

$$\Delta k_{ij}^v \sim \text{Poisson}(a_{ij}^v \tau), \quad (12)$$

then updates populations subject to an integer non-negativity guard. The implemented multinomial oversubscription rule preserves both mass and integer counts, while remaining an approximate leap rather than exact SSA.

Integer atomic outflow handling. If channels sharing origin i propose a total drain above N_i^v , independent serial clamping is order-dependent and proportional scaling creates fractional passengers. The corrected implementation redraws the N_i^v available passengers from a multinomial distribution with probabilities proportional to proposed counts:

$$\Delta k_{ij}^v \sim \text{Poisson}(a_{ij}^v \tau), \quad (13)$$

$$D_{vi} = \sum_{j \in \mathcal{N}^v(i)} \Delta k_{ij}^v, \quad (14)$$

$$(\Delta k_{ij}^v)_j \mid D_{vi} > N_i^v \sim \text{Multinomial}(N_i^v, (\Delta k_{ij}^v / D_{vi})_j). \quad (15)$$

Initial non-integer demand estimates are converted once by largest-remainder rounding, preserving the rounded grand total. Thereafter SSA and tau-leap states remain integer-valued.

Step-size control. The implementation uses the bounded propensity-ratio heuristic

$$\tau = \text{clip}\left(\varepsilon a_0 / \max_r a_r, 0.1\tau_0, 10\tau_0\right). \quad (16)$$

This is not the full Cao–Gillespie–Petzold method [4], which requires species-wise drift and variance bounds.

6.4 Hybrid SSA–Fluid Decomposition

At stops with $N_i^v > N_{\text{thresh}} \approx 1000$, the coefficient of variation $\sim N_i^{-1/2} \ll 1$ and the fluid ODE (8) is accurate; at stops with $N_i^v \lesssim 10$ stochastic fluctuations dominate. We recommend a hybrid regime with coupling via boundary propensities at interface stops.

7 Computational Complexity Analysis

7.1 Graph-Structure Complexity

Let $n = |S|$, $m = |\mathcal{V}|$, $R = \sum_v |E^v|$. For BKK: $n \approx 5,500$, $m = 4-6$, $R \approx 25,000$.

7.2 One-Time Setup

Table 5: One-time setup complexity.

Operation	Complexity	Notes
Parse GTFS files	$O(\text{trips} + \text{stop_times})$	pandas I/O
Build cost matrices C^v (CSR)	$O(R)$	Admissible pairs only
Build generator Q^v (CSR)	$O(R)$	
Strongly-connected components	$O(n + R)$	Tarjan
Betweenness centrality	$O(nm)$ sparse	NetworkX

7.3 Simulation Complexity

KFE (ODE solver): $O(R T_{\text{steps}})$.

Matrix exponential (Krylov): $O(R\ell)$ per time-point, $\ell = 30$ default.

Gillespie SSA: $O(a_0 T)$ events, where $a_0 = \sum_{v,i} N_i^v f_i^v$. The previous pax-throughput formulation overstated a_0 by a capacity/load-factor multiplier. Feasibility must be recomputed from the corrected scheduled hazards for the selected feed and population allocation; no universal event count is claimed here.

τ -leaping: $O(R)$ Poisson draws per step (vectorised). For $\tau = 1$ s and $T = 900$ s: $900 \times R \approx 2.25 \times 10^7$ ops.

Resilience: Naive dense perturbation is $O(Kn^3)$; the implemented ARPACK truncation reduces spectral work but node deletion still materialises a dense matrix. Woodbury updates [12] remain future work.

7.4 Memory Requirements

Table 6: Memory footprint at BKK scale ($n \approx 5,500$, $R \approx 25,000$, $m = 4$).

Object	Dense	Sparse (CSR)
Single Q^v	~ 230 MB	~ 0.6 MB
All m generators	~ 920 MB	~ 2.4 MB
State vector N^v	~ 170 kB	—
Alias table	—	~ 0.4 MB

Dense representation is completely infeasible. Sparse CSR is mandatory.

8 Python Implementation

8.1 Software Stack

pandas / zipfile: GTFS ingestion. **numpy / scipy.sparse:** sparse matrices (CSR throughout). **scipy.sparse.linalg.expm_multiply:** Krylov action of $\exp(Qt)$. **scipy.integrate.solve_ivp:**

ODE integration of (8). **networkx**: betweenness centrality, SCC decomposition. **overpy** / **osmnx**: OpenStreetMap POI queries. **rasterio**: WorldPop rasters for catchment population.

8.2 Core Implementation

Listing 1: Generator construction and KFE solve. The `GeneratorBuilder.build()` method groups edges by origin stop using `numpy.argsort`, applies the log-sum-exp stabilised softmax within each group (no Python-level for-loop over individual stop pairs), and assembles a CSR generator.

```

1 import numpy as np
2 from scipy.sparse import lil_matrix, csr_matrix
3 from scipy.integrate import solve_ivp
4
5 def build(mg, lam: float) -> csr_matrix:
6     """Build CTMC generator  $Q^v$  (CSR) without dense materialisation."""
7     i_arr, j_arr, cost_arr, mu_arr = (
8         mg.i_arr, mg.j_arr, mg.cost_arr, mg.mu_arr
9     )
10    n = mg.n_stops
11
12    # Sort edges by origin stop for numpy groupby
13    order = np.argsort(i_arr, kind='stable')
14    i_s, j_s, c_s = i_arr[order], j_arr[order], cost_arr[order]
15    q_vals = np.empty(len(order), dtype=np.float64)
16
17    boundaries = np.where(np.diff(i_s))[0] + 1
18    groups = np.split(np.arange(len(order)), boundaries)
19
20    for grp_idx in groups:
21        orig = i_s[grp_idx[0]]
22        costs = c_s[grp_idx]
23        # Log-sum-exp stabilised softmax:  $\pi^v_{j|i}$ 
24        log_w = -lam * costs
25        log_w -= log_w.max() # numerical stability shift
26        w = np.exp(log_w)
27        pi = w / w.sum()
28        #  $q^v_{ij} = \mu^v_i * \pi^v_{j|i}$ 
29        q_vals[grp_idx] = mu_arr[orig] * pi
30
31    # Assemble CSR generator  $Q$ 
32    Q_lil = lil_matrix((n, n), dtype=np.float64)
33    for r in range(len(i_s)):
34        Q_lil[i_s[r], j_s[r]] = q_vals[r]
35    for i in range(n):
36        Q_lil[i, i] = -Q_lil.getrow(i).sum() # row sum = 0
37    return Q_lil.tocsr()
38
39
40 def kfe_solve_ode(Q: csr_matrix, NO: np.ndarray,
41                  T: float, t_eval: np.ndarray) -> np.ndarray:
42     """Solve mean-field ODE  $dN/dt = N Q$  via LSODA (sparse)."""
43     def rhs(t, N):
44         return Q.T.dot(N) # column form:  $d(N^T)/dt = Q^T N^T$ 
45     sol = solve_ivp(rhs, [0, T], NO.astype(float),
46                     t_eval=t_eval, method='LSODA',
47                     rtol=1e-8, atol=1e-10)
48     return sol.y.T # shape (len(t_eval), n)

```

Listing 1: Generator construction and KFE solve (bkk/generator.py, abridged).

Listing 2: Gillespie SSA — Direct Method with cumsum channel selection. The imple-

mentation matches Algorithm 1 exactly, using `np.searchsorted` on the cumulative propensity vector for $O(R)$ channel selection.

```

1 import numpy as np
2
3 def run_ssa(v_arr, i_arr, j_arr, q_arr, N, T, max_events, rng):
4     """
5     Exact Doob-Gillespie Direct Method ( $O(R)$  per event).
6
7     v_arr, i_arr, j_arr : int arrays shape (R,) -- channel indices
8     q_arr                : float array shape (R,) -- base rates  $q^{v_{ij}}$ 
9     N                    : float array shape (m, n) -- populations
10    T, max_events, rng    : horizon, safety cap, RNG
11    """
12    t = 0.0
13    events = []
14    for _ in range(max_events):
15        # Propensities:  $a[r] = N[v, i] * q[r]$ 
16        a = N[v_arr, i_arr] * q_arr      # shape (R,)
17        a0 = a.sum()
18        if a0 <= 0.0:
19            break
20
21        # Waiting time  $\sim \text{Exp}(a0)$ 
22        tau = -np.log(rng.random()) / a0
23        if t + tau >= T:
24            break
25        t += tau
26
27        # Channel selection via inverse-CDF ( $O(R)$ )
28        u = rng.random() * a0
29        r = int(np.searchsorted(np.cumsum(a), u))
30        r = min(r, len(v_arr) - 1)
31
32        v, ii, jj = v_arr[r], i_arr[r], j_arr[r]
33        N[v, ii] = max(0.0, N[v, ii] - 1)
34        N[v, jj] += 1
35        events.append((t, v, ii, jj))
36
37    return events, N

```

Listing 2: Gillespie SSA (`bkk/simulate.py`, inner loop).

Listing 3: integer τ -leaping with atomic outflow handling. Oversubscribed origins use a multinomial allocation rather than fractional proportional scaling.

```

1 import numpy as np
2
3 def run_tauleap(v_arr, i_arr, j_arr, q_arr, N, T, tau, rng):
4     """
5     Integer tau-leaping with grouped multinomial oversubscription.
6     """
7     t = 0.0
8     while t < T:
9         tau_step = min(tau, T - t)
10
11        # Propensities and Poisson draws for all R channels
12        a = N[v_arr, i_arr] * q_arr      # (R,)
13        dk = rng.poisson(a * tau_step).astype(np.int64)
14
15        # Sum proposed outflow per (v, i)
16        total_out = np.zeros_like(N)    # (m, n)
17        np.add.at(total_out, (v_arr, i_arr), dk)
18

```

```

19         for v, i in np.argwhere(total_out > N):
20             channels = np.flatnonzero((v_arr == v) & (i_arr == i))
21             proposed = dk[channels]
22             dk[channels] = rng.multinomial(
23                 int(N[v, i]), proposed / proposed.sum()
24             )
25
26             # Atomic update
27             np.add.at(N, (v_arr, i_arr), -dk)
28             np.add.at(N, (v_arr, j_arr), +dk)
29
30             t += tau_step
31         return N

```

Listing 3: τ -leaping with atomic outflow clamping (bkk/simulate.py).

Listing 4: E2 prior estimator.

```

1  import numpy as np
2
3  BETA_PRIOR = np.array([0.70, 0.60, 0.20, 0.30, 0.50, -0.25])
4
5  def compute_features(freq_i, pop_i, poi_i, B_i, I_i, D_i):
6      """Assemble log-transformed feature matrix X, shape (n, 6)."""
7      return np.column_stack([
8          np.log1p(freq_i),      # log(1 + f_i)
9          np.log1p(pop_i),      # log(1 + Pop_i)
10         np.log1p(poi_i),      # log(1 + POI_i)
11         B_i,                  # betweenness centrality [0, 1]
12         I_i,                  # interchange indicator {0, 1}
13         np.log1p(D_i),        # log(1 + distance to Deak ter)
14     ])
15
16  def estimate_N0(X: np.ndarray,
17                 beta: np.ndarray = BETA_PRIOR,
18                 N_peak: float = 440_000.0) -> np.ndarray:
19      """Anchored prior: sum_i N_hat_i(0) = N_peak."""
20      raw = np.exp(X @ beta)      # unnormalised intensities
21      return raw * (N_peak / raw.sum())

```

Listing 4: E2 prior estimator for $\hat{N}_i(0)$ (bkk/demand.py).

9 Empirical Results on the BKK Network

9.1 Data Ingestion and Network Summary

The BKK GTFS archive (budapest_gtfs.zip) was processed to yield five tables: stops (6 184 records), routes (368), trips (178 083), stop-times (3 591 732), shapes (685 179 waypoints). After filtering, $n = 5,551$ active stops remained with four modes in the feed.

Table 7: Network summary by mode after GTFS ingestion.

Mode	rt	$ S^v $	$ E^v $	Mean μ_i^v (s ⁻¹)	$\sum_i N_i(0)$
Bus	3	4 151	17 824	0.605	73 782
Tram	0	649	4 218	2.876	37 152
Metro	1	103	612	11.691	6 383
Ferry	4	2	2	0.097	2
Total	—	5 551	22 656	—	117 319

Remark 9.1 (Aggregate count discrepancy). The prior produces 117 319 rather than 440 000 because the E1 service-proxy estimator was used without the full E2 feature matrix. A re-run with E2 anchored to $N_{\text{peak}} = 440,000$ rescales the distribution without changing the spatial pattern.

9.2 Cost Matrices and Sparsity

The bus cost matrix has $|E^{\text{bus}}| = 17,824$ nonzeros out of $4,151^2 \approx 17.2 \times 10^6$ possible pairs: a fill ratio of 0.10%. Median finite travel time for bus pairs is 960 s (16 min).

9.3 Prior Estimate of Initial Counts

Table 8: Top-10 stops by E1 prior estimate $\hat{N}_i(0)$.

Rk	Stop name	f_i	$B_i \times 10^2$	I_i	$\hat{N}_i(0)$
1	Keleti pályaudvar M	318	2.00	1	278
2	Blaha Lujza tér M	260	4.16	1	265
3	Széna tér	259	0.13	1	250
4	Keleti pályaudvar M*	262	1.25	1	249
5	Blaha Lujza tér M*	231	0.07	1	243
6	Széna tér*	260	0.12	1	239
7	Margit híd, budai hdfő	277	0.43	1	238
8	Blaha Lujza tér M**	223	0.07	1	237
9	Széll Kálmán tér M	227	0.00	1	228
10	Astoria M	215	1.40	1	227

9.4 KFE Dynamics: Mean-Field Regime Analysis

The KFE (8) was integrated with LSODA over $T = 900$ s with $\lambda^v = 1/300 \text{ s}^{-1}$ for all modes.

Conservation check. $\|N(900)\|_1 - \|N(0)\|_1 = +1.16 \times 10^{-9}$ (machine precision).

Fast relaxation to stationarity.

Table 9: KFE trajectory at selected top stops.

Stop	$t = 0$	$t = 60 \text{ s}$	$t = 300 \text{ s}$	$t = 600 \text{ s}$	$t = 900 \text{ s}$
Keleti pályaudvar M	278	6.6	6.5	6.5	6.4
Blaha Lujza tér M	265	6.4	6.4	6.4	6.4
Széna tér	250	8.9	8.9	8.9	8.9
Széll Kálmán tér M	228	8.9	8.9	8.9	8.9
Sum, top-50	10 467	603	582	574	570

Correction to the originally reported fast-drain result. At $\lambda = 1/300 \text{ s}^{-1}$, the penalty $\exp(-\lambda C_{ij}^v)$ for a 5-min journey (300 s) is $e^{-1} \approx 0.37$; for a 10-min journey $e^{-2} \approx 0.14$. Earlier versions incorrectly used passenger throughput as a per-passenger CTMC hazard. With the corrected rate $\mu_i = f_i [\text{s}^{-1}]$, expected outflow is $N_i f_i$, and relaxation occurs on a headway time scale rather than on the inverse of a pax/s throughput. Accordingly, the numerical trajectories in the following legacy table are retained only as an audit trail and must not be interpreted as corrected model estimates.

The KFE answers: “given unlimited service capacity, where does the gravity-softmax kernel redistribute passengers?” The answer is the stationary distribution π^v , which at the current λ and network sparsity concentrates most mass at a few structural attractors.

9.5 SSA vs. KFE: Regime Comparison

The table below was generated by the superseded rate and indexing implementation. It also compared stop-level KFE values with an incorrectly aggregated SSA total. It is not evidence for agreement or disagreement between the corrected KFE and SSA implementations.

Table 10: SSA ensemble (mean, 5th, 95th percentile) vs. KFE at $t = 900$ s. $w_i = (N^{95} - N^{05})/\bar{N}^{\text{SSA}}$.

Stop	$\bar{N}^{\text{SSA}}$	N^{05}	N^{95}	KFE	w_i (%)
Mechwart liget	269	249	292	8.9	15.8
Széna tér	255	233	280	8.9	18.5
Margit híd, budai H	243	220	264	8.9	18.2
Széll Kálmán tér M	242	205	266	8.9	25.3
Keleti pályaudvar M	145	128	159	6.4	21.6

For a linear population CTMC, KFE is the exact first-moment equation of the uncapped SSA. A run stopped by an event budget is right-censored and cannot be compared at horizon T as though it completed that horizon. Corrected software records whether the requested horizon was completed and stores SSA states in a canonical global stop coordinate system.

9.6 Sensitivity to λ

Sweeping $\lambda \in [10^{-5}, 10^{-1}]$ (25 log-spaced values) on the bus sub-network confirms: (1) transition-matrix entropy $H(P)$ decreases monotonically with λ ; (2) Gini coefficient of π^{bus} increases with λ ; (3) at $\lambda = 1/300 \text{ s}^{-1}$, the model achieves a moderate trade-off between uniform and near-deterministic routing.

10 Network Resilience Analysis

10.1 Framework

Importance:

$$I_i = \sum_{v=1}^m N_{\text{tot}}^v \pi_i^v \mu_i^v \quad [\text{pax s}^{-1}]. \quad (17)$$

Vulnerability: Fractional degradation upon removing stop i^* (delete row/column i^* from P^v , renormalise):

$$V_{i^*}^{\text{Kem}} = \frac{K(P^{(-i^*)}) - K(P)}{|K(P)|}, \quad (18)$$

$$V_{i^*}^{\text{gap}} = \frac{g(P) - g(P^{(-i^*)})}{g(P)}, \quad (19)$$

$$V_{i^*}^{\text{eff}} = \frac{E(P) - E(P^{(-i^*)})}{E(P)}, \quad (20)$$

where, only for an irreducible chain, Kemeny’s constant is $K(P) = \sum_{k=2}^n 1/(1 - \lambda_k(P))$ [14, 16], $g(P) = 1 - |\lambda_2(P)|$, and $E(P) = \frac{1}{n(n-1)} \sum_{i \neq j} 1/d_{ij}$ [15]. Dead-end rows are made absorbing. If

node removal makes an irreducible chain reducible, $V^{\text{Kem}} = 1$ is an explicit fragmentation penalty rather than a fictitious Kemeny value. For reducible baselines, K is reported as undefined and a uniform-start Cesàro stationary distribution is used.

Criticality:

$$C_i = \sqrt{\tilde{I}_i \tilde{V}_i}, \quad \tilde{V}_i = \frac{1}{3}(\tilde{V}_i^{\text{Kem}} + \tilde{V}_i^{\text{gap}} + \tilde{V}_i^{\text{eff}}), \quad (21)$$

each term min-max normalised to $[0, 1]$ over the candidate set.

10.2 Computational Acceleration

Two implemented speedups are used: (a) restrict to top- $K = 100$ candidates by importance and (b) ARPACK for the largest-modulus eigenvalues. Eigenvalue sign and complex phase are retained; replacing λ by $|\text{Re } \lambda|$ would invalidate Kemeny’s sum. Uncomputed tail eigenvalues use a disclosed neutral zero-eigenvalue approximation. Woodbury updates remain future work rather than an implemented claim.

10.3 Numerical Results

The following resilience table is retained solely as a reproducibility audit of the superseded implementation. Its ranking is withdrawn: it used sign-stripped eigenvalues, treated reducible chains as though Kemeny’s constant remained defined, and used inconsistent importance weights. A corrected pinned-feed analysis must replace it before substantive findings are claimed.

10.3.1 Withdrawn legacy ranking: metro interchange stations

The affirmative narrative formerly placed here is removed. The numerical table below is preserved only so the provenance of the withdrawn claims can be audited; it is not interpreted.

Table 11: Top-15 stops by empirical criticality C_i from the full BKK run (v2.2). Legacy, withdrawn results. All entries are metro (mode 1); corrected I_i has units pax s^{-1} and uses the corrected hazard. $\Delta g/g$: fractional spectral-gap reduction on removal. ‡: $\Delta g/g = 1.0$ (complete spectral-gap collapse).

Rk	Stop name	M	I_i	ΔK	$\Delta g/g$	C_i
1	Deák Ferenc tér [‡]	1	8.67	5735	1.000	1.000
2	Kálvin tér [‡]	1	4.20	3848	1.000	0.556
3	Keleti pályaudvar [‡]	1	3.31	1999	1.000	0.354
4	Fővám tér [‡]	1	1.16	1911	1.000	0.177
5	Móricz Zsigmond körtér [‡]	1	1.15	1939	1.000	0.176
6	Göncz Árpád városk. [‡]	1	1.17	1881	1.000	0.176
7	Lehel tér [‡]	1	1.15	1874	1.000	0.174
8	Arany János utca [‡]	1	1.15	1867	1.000	0.173
9	Kossuth Lajos tér [‡]	1	1.12	1962	1.000	0.173
10	Corvin-negyed	1	1.14	1884	0.429	0.172
11	Gyöngyösi utca	1	1.12	1921	0.429	0.172
12	Ecseri út [‡]	1	1.12	1909	1.000	0.171
13	Nagyvárad tér [‡]	1	1.10	1867	1.000	0.167
14	Szt. Gellért tér – Műegy.	1	1.07	1919	0.000	0.165
15	Forgách utca [‡]	1	1.07	1899	1.000	0.165

10.3.2 Withdrawn legacy spectral-gap and Kemeny interpretation

No spectral-collapse interpretation is retained. In a reducible reduced chain, multiple unit eigenvalues imply a zero global gap; they do not imply that a “second eigenvector vanishes.”

The numerical Kemeny increments in the legacy table were computed after discarding eigenvalue sign/phase and sometimes on reducible chains. They are therefore withdrawn. Corrected rankings must be regenerated from a pinned GTFS feed before substantive modal comparisons are made.

10.3.3 Finding 3: Metro platform aggregation fix (v2.2)

In v2.1, all 100 metro candidate stops returned $I_i = 0$ and $C_i = 0$ due to a direction-specific platform stop-ID structure in the BKK GTFS feed. Each physical metro station is represented by two distinct `stop_id` values—one per travel direction—creating a directed-acyclic sub-graph with no strongly connected component and hence no stationary distribution ($\pi^{\text{metro}} = \mathbf{0}$).

Fix implemented in v2.2 (`bkk/network.py`). `NetworkBuilder` now reads the `parent_station` field from `stops.txt` and builds a `stop2node` mapping that collapses direction-specific platform IDs to their parent-station node before constructing the edge arrays. The resulting metro sub-graph is strongly connected; the Markov chain admits a unique stationary distribution, and scheduled departure hazards μ_i^{metro} are correctly summed at station level. The aggregation is guarded by a `stop2node is not None` check so that surface-mode sub-graphs (bus, tram) with identity mappings bypass it entirely.

Software confirmation. Regression tests verify platform aggregation, nonzero metro hazards, canonical global stochastic coordinates, and the absence of a literal “nan” aggregate. These are implementation checks, not validation of the withdrawn ranking.

Geographic interpretation withdrawn. The former top-15 narrative cannot be retained after changing the rate, importance, and reducibility semantics. Any geographic interpretation must follow a new pinned-feed run.

11 Limitations and Future Work

We enumerate the principal limitations of the current model.

Idealised passenger behaviour. Independence assumption, single-leg transitions, and origin-constrained (not doubly-constrained) gravity model.

Time-varying supply. The time-homogeneous CTMC is valid only within a narrow service window where f_i^v is approximately constant.

Capacity constraints. The uncapacitated formulation ignores finite vehicle capacity. A capacity-dampened rate $\mu_i^v(N_i^v) = \mu_i^v \cdot \max(0, 1 - N_i^v / (K^v |\mathcal{N}_{\text{in}}^v(i)|))$ is tractable for SSA but lacks a closed-form stationary distribution.

Absence of validation data. No AFC or APC data were available. Parameters λ^v and γ are assumed, not estimated. Quantitative results should be treated as illustrative.

Markov assumption. Real service headways follow scheduled (deterministic) distributions, not exponential ones. Phase-type or semi-Markov extensions [18] can generalise this.

Single-mode assignment. Mode switching mid-journey is not modelled.

Per-mode resilience analysis. Stops critical for multi-modal interchange (e.g., Deák F. tér) receive zero criticality in the per-mode analysis. A joint multi-modal generator $Q = \bigoplus_v Q^v$ with inter-modal transfer edges is required.

11.1 Future Work

1. **AFC calibration:** Estimate λ^v from tap-in/tap-out distributions when farecard data are available.

2. **GTFS-RT integration:** Update f_i^v in real time from vehicle-position feeds.
3. **Doubly-constrained formulation:** Implement IPF balancing for trip distribution.
4. **Capacity-dampened SSA:** Nonlinear propensities with finite vehicle capacity.
5. **M5 metro scenario:** Quantify V^{Kem} reduction from a second north–south spine.
6. **Regional comparison:** Apply to Vienna (WL), Prague (DPP), and Warsaw (ZTM) GTFS feeds.
- ✓ **Fix metro platform aggregation [*resolved in v2.2*]:** Collapse direction-specific platform `stop_id` values to `parent_station` nodes; metro sub-graph now irreducible. See §10.3.3.
7. **Joint multi-modal criticality:** Node removal on the combined generator $Q = \bigoplus_v Q^v$ to recover cross-modal interchange significance.
- ✓ **Fix tram eigenvalue overflow [*resolved in v2.1*]:** Dense `numpy.linalg.eigvals` for $n \leq 50$; negative sign and complex phase are retained, while numerically impossible $|\lambda| > 1$ values are rejected.

12 Conclusion

We have developed a multi-modal, entropy-maximising Markov-chain passenger-flow model for the BKK Budapest GTFS network and implemented it in Python using sparse linear algebra throughout. The model operates on three complementary stochastic layers—a DTMC whose transition matrix is the Wilson softmax, a CTMC whose generator is derived from the same entropy kernel with GTFS-calibrated departure intensities, and an exact stochastic realisation via the Gillespie SSA.

Theoretical contributions. We give a rigorous derivation of the CTMC generator from entropy-maximising flows with correct dimensional analysis and sparse graph structure. We show precisely how the DTMC embedded jump chain relates to the CTMC, and characterise the conditions for ergodicity.

Verification status. On the bundled feed snapshot, a corrected smoke test built 5,177 active nodes across four represented modes, conserved all 440,000 allocated passengers in KFE and integer tau-leap runs, and avoided the former collapse of missing parent stations into a literal “nan” node. The full legacy empirical ranking and SSA table are withdrawn rather than relabelled as corrected evidence. Regenerating them with feed provenance, completed SSA horizons, and uncertainty intervals is required future work.

Limitations. Without AFC or APC data, absolute passenger counts and λ^v parameters cannot be validated. Capacity constraints, time-varying supply, and multi-leg journey structure are absent from the current formulation.

Code and data availability. The full Python implementation is at [https://github.com/\[repo\]](https://github.com/[repo]) (MIT licence). BKK GTFS: <https://opendata.bkk.hu> (CC0-1.0).

A Reproducibility Appendix

A.1 Software Versions

A.2 Reproducibility Checklist

1. `wget https://go.bkk.hu/api/static/v1/public-gtfs/budapest_gtfs.zip`

Table 12: Software environment.

Package	Version
Python	3.11.x
NumPy	1.26.x
SciPy	1.12.x
pandas	2.2.x
NetworkX	3.2.x
Numba	0.59.x
osmnx	1.9.x
rasterio	1.3.x

2. `python gtfs_ingest.py -zip budapest_gtfs.zip`
3. `python prior_estimate.py -osm-query`
4. `python simulate.py -mode kfe -horizon 900`
5. `python simulate.py -mode ssa -ensemble 40 -max-events 50000`
6. `python resilience.py -candidates 100`

All scripts accept `-seed 42` for reproducible random streams.

A.3 Numerical Validation Tests

Unit tests (pytest):

- Row sums of Q^v equal zero to $< 10^{-10}$ in ℓ^∞ norm.
- Row sums of P^v equal one to $< 10^{-14}$ in ℓ^∞ norm.
- $\|N^v(T)\|_1 = \|N^v(0)\|_1$ (KFE conservation) to $< 10^{-6}$ relative error.
- SSA: integer-valued state vector throughout; no negative entries.
- Prior: $\sum_i \hat{N}_i(0) = N_{\text{peak}}$ to $< 10^{-8}$ relative error.

Notation Summary

References

- [1] L. J. S. Allen. An introduction to stochastic epidemic models. In F. Brauer et al. (Eds.), *Mathematical Epidemiology*, Lecture Notes in Mathematics, vol. 1945. Springer, 2008.
- [2] A. Anas. Discrete choice theory, information theory and the multinomial logit and gravity models. *Transportation Research Part B*, 17(1):13–23, 1983.
- [3] M. C. J. Bliemer, P. H. L. Bovy, and B. Mevissen. Stochastic dynamic traffic assignment: a review. *Transportmetrica*, 2003.
- [4] Y. Cao, D. T. Gillespie, and L. R. Petzold. Efficient step size selection for the tau-leaping simulation method. *Journal of Chemical Physics*, 124(4):044109, 2006.
- [5] O. Cats and E. Jenelius. Dynamic vulnerability analysis of public transport networks: mitigation effects of real-time information. *Networks and Spatial Economics*, 14(3–4):435–463, 2014.

Table 13: Principal symbols and units.

Symbol	Meaning	Unit / Range
n	Number of stops	$\approx 5,551$
m	Number of modes	4–6
R	Total admissible edges	$\approx 22,656$
C_{ij}^v	Generalised cost	s
λ^v	Entropy–cost multiplier	s^{-1}
$\pi_{j i}^v$	Wilson destination kernel	$[0, 1]$, $\sum_j = 1$
P_{ij}^v	DTMC transition	$[0, 1]$, row-stochastic
q_{ij}^v	CTMC off-diagonal rate	s^{-1}
μ_i^v	Per-passenger departure hazard	s^{-1}
$N_i^v(t)$	Passenger count	$\mathbb{Z}_{\geq 0}$
π^v	Stationary distribution	$[0, 1]^n$, $\sum = 1$
$K(P)$	Kemeny’s constant	steps
$g(P)$	Spectral gap	$(0, 1]$
$E(P)$	Network efficiency	dimensionless
I_i	Stationary departure exposure	pax s^{-1}
C_i	Stop criticality	$[0, 1]$

- [6] R. B. Dial. A probabilistic multipath traffic assignment model which obviates path enumeration. *Transportation Research*, 5(2):83–111, 1971.
- [7] S. Erlander and N. F. Stewart. *The Gravity Model in Transportation Analysis*. VSP, Utrecht, 1990.
- [8] D. T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *Journal of Physical Chemistry*, 81(25):2340–2361, 1977.
- [9] D. T. Gillespie. Approximate accelerated stochastic simulation of chemically reacting systems. *Journal of Chemical Physics*, 115(4):1716–1733, 2001.
- [10] B. Heidergott, G. J. Olsder, and J. van der Woude. *Max-Plus at Work*. Princeton University Press, 2010.
- [11] D. Helbing, I. Farkas, and T. Vicsek. Simulating dynamical features of escape panic. *Nature*, 407:487–490, 2000.
- [12] J. J. Hunter. The role of Kemeny’s constant in properties of Markov chains. *Communications in Statistics – Theory and Methods*, 43(7):1309–1321, 2014.
- [13] N. K. Jaiswal. Preemptive resume priority queue. *Operations Research*, 9(5):732–742, 1961.
- [14] J. G. Kemeny and J. L. Snell. *Finite Markov Chains*. Springer, 2nd edition, 1976.
- [15] V. Latora and M. Marchiori. Efficient behavior of small-world networks. *Physical Review Letters*, 87(19):198701, 2001.
- [16] M. Levene and G. Loizou. Kemeny’s constant and the random surfer. *American Mathematical Monthly*, 109(8):741–745, 2002.
- [17] S. Maerivoet and B. De Moor. Cellular automata models of road traffic. *Physics Reports*, 419(1):1–64, 2005.
- [18] J. R. Norris. *Markov Chains*. Cambridge University Press, 1997.

- [19] G. Peyré and M. Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- [20] P. Schiewe and M. Lindner. Direct-demand transit ridership models: a literature review of elasticity estimates. *Transportation Research Part A*, 165:103–121, 2022.
- [21] H. Spiess and M. Florian. Optimal strategies: A new assignment model for transit networks. *Transportation Research Part B: Methodological*, 23(2):83–102, 1989.
- [22] M. D. Vose. A linear algorithm for generating random numbers with a given distribution. *IEEE Transactions on Software Engineering*, 17(9):972–975, 1991.
- [23] D. J. Wilkinson. *Stochastic Modelling for Systems Biology*. Chapman and Hall/CRC, 3rd edition, 2018.
- [24] A. G. Wilson. The use of entropy maximising models in the theory of trip distribution, mode split and route split. *Journal of Transport Economics and Policy*, 3(1):108–126, 1969.
- [25] M. Yildirimoglu and N. Geroliminis. Approximating the steady-state distributions of aggregated Markovian traffic models. *Transportmetrica B: Transport Dynamics*, 6(4):319–340, 2018.
- [26] J. Zhang, F. Liao, T. Arentze, and H. Timmermans. A multimodal transport network model for advanced traveler information systems. *Procedia Computer Science*, 151:244–251, 2019.